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Research paper

Sizing of Multirotor Aerial Vehicle: A Statistical Approach

Aditya Thokur Ravi [a], Manan V Jain [a,b], Sashank Erukala [a], Vallabhu Narayana Murthy [a], Vishwa Karthi [a], Narendiran V [a], Rajesh Senthil Kumar T [a,*]

[a] Department of Aerospace Engineering, Amrita School of Engineering, Coimbatore, Amrita Vishwa Vidyapeetham, India
[b] TUD Dresden University of Technology, Faculty of Computer Science, 01187, Dresden, Germany

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ABSTRACT

Multirotor Unmanned Aerial Vehicles (UAVs) serve a wide range of applications, from micro-scale inspection to heavy-lift industrial systems, yet the preliminary design process lacks a systematic, scalable methodology. This study establishes a class-specific, statistically derived framework for sizing multirotor UAVs. A dataset of 309 commercially available UAVs is compiled and classified into two classes based on Maximum Take-Off Weight (MTOW): Class I includes 179 UAVs with MTOW less than 1 kg, and Class II includes 130 UAVs with MTOW greater than 1 kg. The correlations between the MTOW and design variables are derived for each class. These relations show distinct scaling behaviors. Class I shows sublinear geometric scaling whereas Class II exhibits near-linear or super-linear scaling. The observed relations are used to formulate a sequential sizing methodology. A case study is performed to validate the proposed methodology. In the case study, a heavy-lift UAV for 60 kg payload is designed using the derived scaling laws and its dynamic stability characteristics are evaluated using MATLAB/Simulink. Stable trajectory tracking performance confirms the dynamic feasibility of the UAV sized using the proposed framework. Overall, this study forms a statistical design framework for multirotor sizing and provides estimates, serving as a starting point for early-stage design, which is to be followed by physics-based design, mission-specific trade-offs among parameters such as endurance and range, and optimization in subsequent stages.

1. Introduction

Multirotor Unmanned Aerial Vehicles (UAVs) have grown from simple hobbyist flying machines to indispensable assets across a wide range of civil and defense applications. Their vertical take-off and landing capability, maneuverability, and scalability across different weight classes make UAVs suitable for a range of missions[1,2] and applications, such as agricultural management, aerial imaging, geo-spatial mapping, logistics, disaster relief, military reconnaissance, and warfare operations. Applications of multirotor exist across a wide operational spectrum, from less than 250g nano aerial vehicles (NAV) used for structural inspections and indoor mapping [1,3], to more than 50kg heavy lift UAVs able to carry industrial payloads, firefighting equipment, or modular sensor suites for environmental monitoring [2,4]. These extreme application ranges pose fundamentally different design regimes, emphasizing the need for tailored design requirements, corresponding to mission priorities. To account for these aspects, designers must balance coupled factors such as propulsion efficiency, structural mass, payload capacity, endurance, and battery capacity, yet no universally accepted standardized guidelines exist. In contrast, the classical fixed-wing aircraft design method benefits from well-established principles of design based on performance equations, semi-empirical, statistical, and analytical frameworks [5–9].

With the rise of electric propulsion in the mid-2000s, attention turned to sub-1kg UAVs. Bouabdallah et al. [3] demonstrated a propulsion-group design methodology for sub-kilogram VTOL platforms, applying combined momentum and blade element theory to achieve hover flight with up to 30 min autonomy and simulated obstacle avoidance in cluttered indoor environments. Winslow et al. [10] combined BEMT with empirical component mass relationships to size rotors, motors, ESCs, batteries, and airframe for quadcopters with gross takeoff weights between 30 and 1000 g, however the method is constrained to the low Reynolds number regime characteristic of micro-scale platforms. Budinger et al. [11] proposed dimensional based scaling laws and similarity models for multirotor drone components such as motors, ESCs, batteries, propellers, and structure and validated against platforms

* Corresponding author.
E-mail address: t_rajesh@cb.amrita.edu (R.S.K. T).

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ranging from ~1.6 kg quadrotors to ~8.5 kg octorotors. On the other end of the spectrum, i.e., the heavy-lift multirotors, Kotarski et al. [2] studied on propulsion units, characterizing thrust, torque, and power drawn sizing charts for multirotors capable of lifting payloads more than 20 kg, and Ong et al. [12] presented coaxial-rotor heavy-lift multirotors for compact urban logistics UAVs.

Recent works focuses primarily on UAV dynamics and control rather than systematic design and sizing [13]. Examples include agricultural UAVs,[14] solar and hybrid quadcopters,[15,16] reinforcement learning-based control[17], morphing and spherical platforms,[18,19] and bioinspired design studies [20,21]. Several studies have examined component-level or subsystem-oriented design. de Angelis et al. [9] applied an analytical framework to electric multirotors, evaluating hovering performance and deriving optimal battery-to-takeoff-weight ratios, though the approach is limited to hover conditions and specific propeller datasets. Biczyski et al. [22] developed a database-driven methodology that selects propeller and motor combinations based on operating points, estimating hovering power, ESC current rating, and battery discharge to produce a preliminary bill of materials with flight-time predictions, though the approach relies on a static hover model, does not capture dynamic flight performance and remained limited to component-level estimations. In parallel, Bershadsky and Haviland [23] developed the Electric Multirotor Sizing Tool (EMST), using parameterized propulsion characteristics such as torque, KV rating, propeller diameter, and battery internal resistance. EMST works on propulsion sizing to meet target thrust and endurance levels [23]. However, airframe mass, structural factors, and system-level correlations remained external inputs, limiting EMST as a full sizing tool.

Delbecq et al. [24] integrated component-level scaling laws governing propellers, motors, ESCs, batteries, and structural parts, into a multidisciplinary design optimisation (MDO) framework, enabling efficient preliminary sizing of electric multirotors across a range of mission requirements and payload configurations. Lorenz's[25] compilation of thermal-powered rotorcraft provided power-law regressions between rotor radius, MTOW, and installed power. Omri & Rand[26] supplemented the dataset and included non-dimensional similarity coefficients. These early studies by Lorenz[25] and Omri & Rand[26], established fundamental geometric and power correlations, but predominantly to single-rotor helicopters and autogyros powered by piston or turboshaft engines. Filippone et al. [31] conducted a statistical scaling study on a population of 93 unmanned wingless rotorcraft, establishing allometric regression relations between MTOW and size and performance parameters across rotor configurations and propulsion types, extending earlier manned-rotorcraft analyses by Lorenz [25] and Filippone[27].

Multirotor design methodologies remain fragmented across the weight classes,[28] oscillating between heuristic rules,[22,29] component level optimizations,[29,30] and limited statistical models [29–31]. Machine learning and genetic algorithm face issues due to computational costs, data scarcity, limited generalizability due lack of physical interpretability, and minimal real-world validation beyond simulation-based results [4,12,32,33]. Other studies in multirotor sizing involve emphasis on optimizing the propulsion system comprising power-train modeling of the propulsion system including the propeller, motor, Electronic Speed Controller, and battery [1,10,11,23,24,30, 32–35]. Moreover, the literature lacks standardized terminology, publicly available datasets, inconsistent rules of thumb for initial assumptions and matching of components instead of performance-based design methodologies. Although statistical methods can facilitate conceptual design, particularly in early-stage sizing and trade-off studies, such approaches remain limited in the multirotor UAV domain.

Existing methods often generalize across scales. The effect of scale is particularly evident in mass distribution, for instance, the structural mass fraction and energy-storage efficiency change drastically with scale: micro UAVs generally carry 30-40% of their mass in electronics and control units,[11] while in heavy-lift systems, structural reinforcement and large propulsion units may constitute over 60% of total mass [2]. A particular issue in the literature is the variability of scaling exponents. While power-law scaling has proved effective within narrow weight bands, the exponents themselves are not universal constants. Comprehensive design perspectives and classification reviews also highlight the need for unified multirotor frameworks [20,21].

Although prior studies have reported statistical scaling relationships, [25,31] they have been used to describe trends rather than providing structured preliminary sizing framework for multirotor UAVs. In early design stages, however, geometric, propulsion and other parameters are often unavailable, and designers rely on these estimates. This creates a need to interpret statistical relations not just as descriptive regressions but a class-aware framework which can support preliminary sizing. Accordingly, this study derives regression relations of commercially available multirotor UAV data to develop a class based preliminary scaling and sizing framework. Rather than limiting the analysis to regression, the objective is to propose a framework that quantitatively relates sizing parameters to design metrics such as MTOW, rotor diameter, battery capacity, and endurance, such that they can be used in a sizing framework to obtain estimates across different weight regimes. However, a single global model across all weight scales is unlikely to give meaningful correlations due to the non-uniform scaling behavior of multirotors across size regimes. To address this, the study introduces an MTOW-based classification. UAVs are categorized into two classes according to their MTOW. Class I includes UAVs with MTOW less than 1 kg, and class II includes UAVs with MTOW greater than 1 kg. The MTOW threshold for classification is motivated by the market segmentation. UAVs below 1 kg are usually compact, integrated sensing platforms in which endurance and battery capacity govern design. Whereas, heavier multirotors carry separate mission payloads and are designed around payload capacity, actuator area, and battery mass. A nested model F test was used for the regression between MTOW and characteristic UAV size, and the class segmented model had a significantly better fit ($F(2,248) = 12.91$, $p < 0.001$), supporting the structural change at this MTOW. The 1kg MTOW threshold ensures that the regression study carried out remains meaningful in its respective UAV classes having similar use case. The study is structured around the following steps:

1. A dataset of publicly available specifications of commercial multirotor UAVs, including MTOW, rotor diameter, battery capacity, frame size, flight time, and payload, is collected.
2. Classification of the dataset based on maximum takeoff weight (MTOW), which is a key factor influencing UAV design.
3. To obtain correlations between MTOW and various design parameters using regression within each class.
4. To bring out a scaling and sizing methodology for multirotors based on commercially available UAV data, enabling preliminary design across different UAV classes.

The paper is organized as follows: Section 2 presents the database formation for both class I and class II multirotors. The data set formation methodology, number of entries, and preliminary statistical characterization are discussed in detail. Section 3 formulates the scaling and sizing relationships by identifying the correlations between MTOW and the design parameters for multirotor sizing. Section 4 applies the proposed methodology to a representative case study. A multirotor UAV is designed using the derived relations and subsequently validated through MATLAB/Simulink-based simulations.

2. Survey & Database Formation

The formulation of scaling relationships and sizing methodology for multirotor UAVs necessitates a comprehensive dataset encompassing the spectrum of commercially available multirotors. This study's primary data collection targeted commercially available off-the-shelf multirotors through a systematic survey of open-source manufacturer specifications,

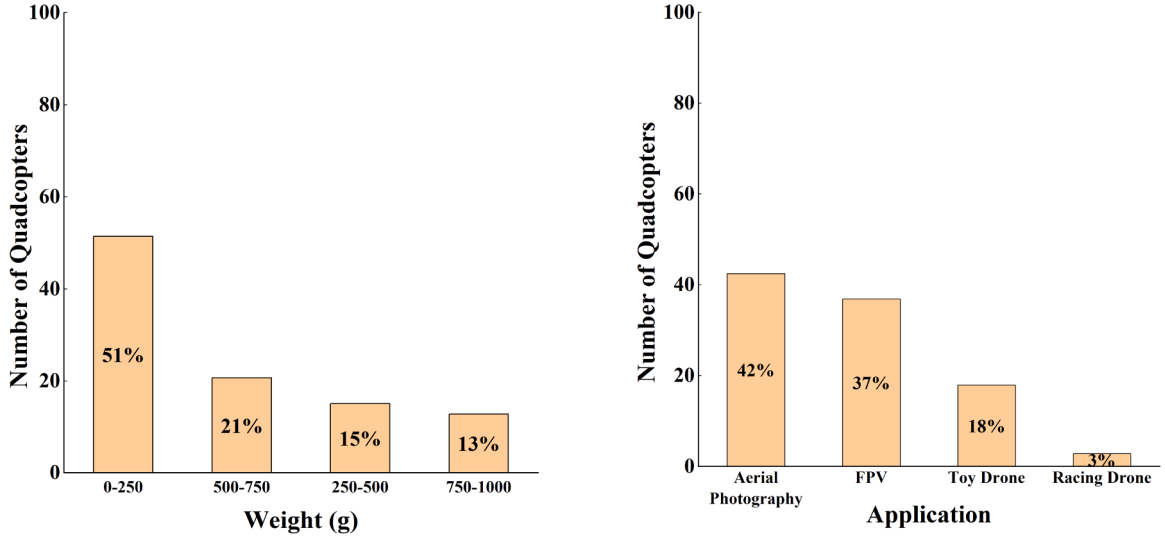
e-commerce platforms, and peer-reviewed databases. Filippone's electric multirotor database was used as initial reference [31]. However, Filippone's database focuses on a smaller set of UAVs, which introduces distribution bias and limits class-wise statistical inference. Hence, a collection of parametric and supplemental data from additional commercial UAVs across all weight categories through direct survey from manufacturer websites and verified e-commerce listings was conducted. The consolidated dataset comprises of 309 unique multirotor UAVs after de-duplication, spanning major commercial weight categories. Building upon the MTOW-based classification framework, the dataset is bifurcated into class I, comprising 179 UAVs with MTOW of less than 1 kg, and class II, consisting of 130 UAVs with MTOW greater than 1 kg. The class-wise multirotor UAVs dataset formed and used in this study is presented as supplementary material. These mass ranges align with common industry categories: micro/light commercial platforms and larger industrial multirotors. Hybrid VTOL configurations (e.g., tilt-rotor or tilt-wing) were excluded, as the lift generation in forward flight is partly due to aerodynamic wing contribution which affect scaling behavior.

2.1. Class I Database

The multirotor distribution across weight classes within class I is shown in Fig. 1a. A pronounced clustering is observed in the sub-250g range i.e., nano-quads. Nano-quads are usually used for imaging, FPV, and recreational applications, having compact, integrated architectures, which are manufactured using standardized components, leading to a concentration of similar configurations in this lower MTOW segment. While, heavier multirotors are designed for mission-specific payload, so they have variability in geometry, actuator area, and thus, overall mass. Although nano quads carry minimal payload and have decent endurance, their affordability, fewer regulatory requirements, mass production and ease of operation make them dominate the commercial market. The 250g threshold, below which many aviation regulations waive registration requirements for, is based on the impact energy estimate of a drone in free fall, and the associated lethality risk of the strike[62]. However, this regulatory boundary does not correspond to a shift in design characteristics, the same is observed in the design trends shown in Section 3. The distribution of commercial applications among class I multirotors reveals four dominant use cases, as shown in Fig. 1b: Aerial Photography (AP), First-Person View (FPV), recreational use such as Toy Drones (TD), and Racing Drones (RD). Consequently, the scaling laws derived in this study will heuristically apply mostly to these dominant application areas considered.

From the compiled dataset, consistently reported design and performance metrics suitable for statistical modeling were identified. Among all the relevant parameters, ten parameters were reliably provided across different manufacturers and deemed appropriate for inclusion in this study. Table 1 shows the statistical characterization of the design and performance metrics for class I dataset. The parameter set encompasses geometric parameters such as frame length "L_I (mm)", breadth "B_I (mm)", and wheelbase "G_I (mm)"; propulsion specifications including propeller diameter "D_I (inch)"; performance metrics such as endurance "E_I (min)", maximum velocity "V_{maxI} (m/s)" wind speed resistance "V_{wI} (m/s)", and service ceiling "H_I (m)"; and mass-energy parameters including maximum takeoff weight "$MTOW_I$ (g)" and battery capacity "BCap (mAh)".

The data availability of a few parameters varies widely due to inconsistent manufacturer reporting practices. For example, wind resistance and service ceiling were reported the least in the collected dataset, reflecting the lack of standardization in industry specifications. Additionally, nano UAVs are typically not designed for carrying payloads. Notably, payload capacity specifications were absent from manufacturer data or seldom reported. This is consistent with their intended use, which usually involves built-in sensors, integrated payloads rather than modular payloads. While MTOW inherently incorporates payload mass, its explicit exclusion from analysis reflects industry reporting practices rather than technical irrelevance. MTOW, however, is consistently available and serves as a key parameter in this study. The propeller diameter and wheelbase were provided in only 28.5% and 30.2% of the entries, respectively. Data that are not available for wheelbase and propeller diameter were estimated following the geometric method proposed by Quan,[36] as illustrated in Fig. 2. The wheelbase and propeller diameter values used for regression therefore include both reported manufacturer data and the estimated values obtained using the geometric relations, equation 1 and equation 2 taken from Quan[36], where G is the wheelbase (mm), D_I the propeller diameter (in), L_I the frame length (mm), and B_I the frame breadth (mm). Equation 2 has a factor of 1.2 to account for typical frame clearances and mounting offsets, and 25.4 is a unit conversion factor from millimeters to inches. The estimated values were compared to available manufacturer data, showing only minor deviations and confirming the validity of



(a) Distribution across weight brackets. (b) Distribution across commercial applications

Fig. 1. Weight and application classifications of class I dataset multirotor UAVs.

Table 1
Statistical characterization of class I UAVs.

Parameters	Notation	Units	Min	Median	Max	Mean	SD	Entries
MTOW	$MTOW_I$	g	17.80	250.00	1000.00	367.92	278.21	179
Length	L_I	mm	25.40	249.00	550.00	242.14	103.34	167
Breadth	B_I	mm	25.40	230.00	500.00	222.32	98.98	166
Propeller Diameter	D_I	inch	1.22	5.00	10.00	5.06	2.90	36
Endurance	E_I	min	2.00	18	90.00	19.85	13.34	157
Wheelbase	G	mm	65.00	231.00	500.00	246.77	122.28	55
Maximum Velocity	$V_{max\ I}$	m/s	6.00	16.00	57.78	21.04	14.69	55
Battery Capacity	BCap	mAh	120.00	1500.00	8000.00	1985.45	1718.30	161
Ceiling Altitude	H_I	m	2000.00	5000.00	7000.00	4827.57	1188.99	37
Windspeed Resistance	V_{wI}	m/s	4.00	11.32	27.78	12.67	4.70	35

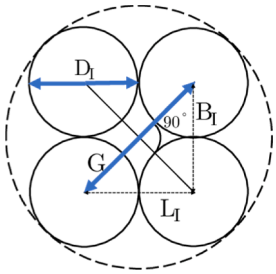


Fig. 2. Schematic of Idealized Quadcopter showing different sizing parameters.

the approximation.

$$G = \sqrt{L_I^2 + B_I^2} \tag{1}$$

$$D_I = \frac{G \cdot \sin(45^\circ)}{1.2 \cdot 25.4} \tag{2}$$

2.2. Class II Database

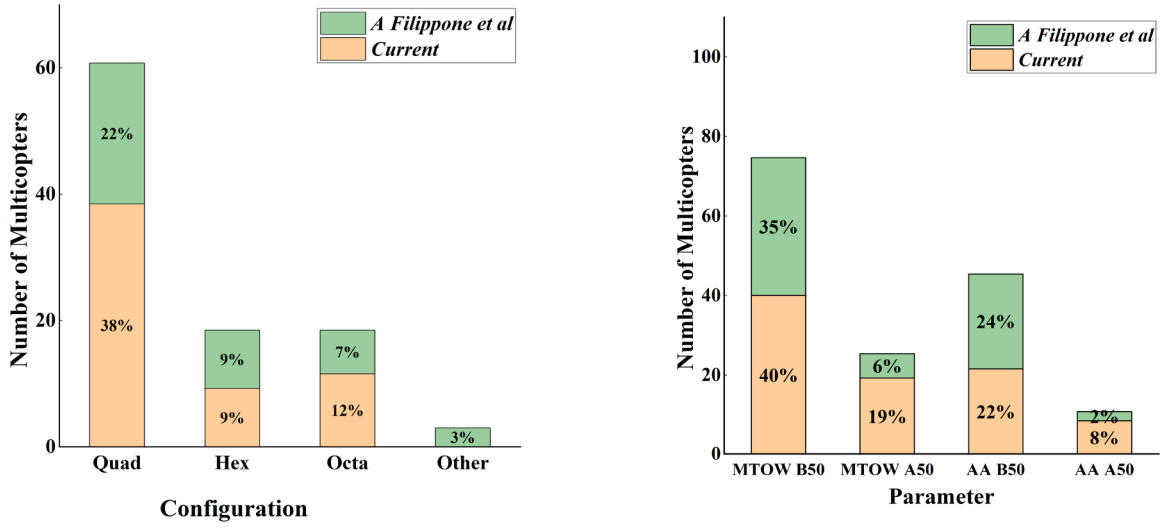
The class II database includes 130 multirotor UAVs with MTOW greater than 1 kg, representing heavier commercial multirotors designed for applications distinct from class I multirotors. Filippone et.al[31] electric multirotor database, while valuable, underrepresents heavier multirotor categories, particularly those exceeding 50 kg MTOW or designed with large actuation areas (AA), critical for contemporary commercial applications. A survey of commercially available off the shelf (COTS) multirotors was conducted to include multirotors exceeding 50 kg MTOW in the class II database. UAVs in this class are primarily used for applications such as agriculture, surveillance missions, aerial surveying, and package delivery, i.e., sectors that demand higher payload capacity, longer endurance, and heavy-duty operating capabilities.

Fig. 3a shows the distribution of multirotor configurations within the Class II dataset. The quadrotor is the dominant configuration, due to their commercial viability and widespread usage. Quadrotors account for 80 out of 130 UAVs in the database, with 51 entries newly added, compared to 29 reported in Filippone et.al[31] dataset. Hexacopters consists of 23 entries, with 12 from Filippone et.al[31] and 11 newly added. There is a notable increase in Octocopter entries, from 9 to 24, indicating a trend towards higher thrust configurations appropriate for agricultural spraying and logistics applications.

Fig. 3b presents a comparative analysis of MTOW and actuation area (AA) within the Class II database. The number of UAVs with MTOW more than 50 kg has increased from 8 to 32. Additionally, the number of multirotors with an actuation area exceeding 50 units has increased to fivefold, from 3 to 16 entries. The actuator area represents the total disk area available for lift generation is computed as:

$$AA = \pi \left(\frac{D}{2}\right)^2 N_P \tag{3}$$

where "D" is the rotor diameter and "N_P" is the number of propellers. This dataset reflects the current industrial trend towards heavier and



(a) Configuration grouping (b) Parameter grouping

Fig. 3. Classifications of multirotor UAVs in the Class II database.

Table 2
Statistical characterization of class II UAVs.

Parameter	Notation	Unit	Min	Median	Max	Mean	SD	Entries
MTOW	$MTOW_{II}$	kg	1.38	20	800.00	55.33	94.87	130
Payload mass	W_p	kg	0.19	10.00	225.00	31.08	47.18	97
Actuation Area	AA	m^2	0.21	6.03	193.02	13.52	26.06	73
Length	L_{II}	m	0.115	1.24	5.61	1.627	1.30	88
Breadth	B_{II}	m	0.078	1.01	22.10	1.63	2.50	85
Range	S_{II}	km	0.80	6.50	110.00	13.97	17.25	70
Endurance	E_{II}	min	7.00	35.00	240.00	40.90	28.90	122
Ceiling Altitude	H_{II}	km	0.015	4.50	7.00	3.51	2.24	68
Cruise Velocity	V_{II}	km/h	1.24	54.00	130.00	56.32	25.58	103
Battery Mass	W_B	kg	0.30	2.80	20.00	4.58	4.91	48
Battery Energy	En_{II}	Wh	15.84	566	4560	961.077	1060.20	67

energy-intensive UAVs, designed for specialized roles and capable of operating under increased mission loads.

Table 2 shows the statistical characterization of the design and performance metrics for the class II data set. The parameter set encompasses geometric parameters such as frame length "L_{II} (m)", breadth "B_{II} (m)"; performance metrics such as endurance "E_{II} (min)", cruise velocity "V_{II} (km/h)", Range "S_{II} (km)" and ceiling altitude "H_{II} (km)"; mass-energy parameters including maximum takeoff weight "$MTOW_{II}$ (kg)", payload mass "W_P (kg)", battery mass "W_B (kg)"; and aerodynamic & sizing indicator such as actuation area "AA (m^2)". Payload mass "W_P" remains under-reported despite its operational significance. Battery energy "En_II (Wh)" is not consistently reported in UAV manufacturer specifications, with only 67 of the 130 Class II UAVs providing this parameter. To supplement these, commercially available lithium-polymer batteries from MaxAmps [37] and GenX [38] were surveyed across the corresponding battery mass range, contributing additional energy-mass data points and bringing the total to 116 entries.

The MTOW varies from 1.05 kg to 800 kg, with a mean of 55.33 kg and a median of 20 kg, highlighting a broad range with a right-skew. This suggests the presence of a small number of very heavy UAVs which disproportionately elevate the average, while most entries are clustered around much lower MTOW values. This uneven density of data introduces statistical imbalance, where a small number of heavy-lift systems disproportionately influence on regression fits, potentially distorting scaling trends. To mitigate such effects, the dataset is segmented at 50 kg MTOW.

3. Scaling Law Derivation and Multirotor Sizing

This section discusses about the scaling and sizing of class I & class II multirotors. The scaling relationships are derived from statistical analysis of the multirotor database formed in the Section 2. The collected parameters are categorized into primary and dependent variables. The primary variable is taken as the MTOW, a readily available, physically meaningful and design constraint parameter. The dependent variables include performance and geometry related quantities such as endurance, battery capacity, wheelbase, propeller diameter, actuation area, length, and breadth. These dependent parameters reflect the performance objectives and physical layout of the UAV. The dependent parameters are expressed in terms of the primary variable using a power law relation regression. The power law model is considered because multirotor parameters follow nonlinear allometric trends, in which geometric and performance quantities do not vary linearly with vehicle size [25,26,31]. Initial power law regression is performed on the complete dataset, followed by outlier removal within a 2σ confidence interval, to reduce the influence of extreme or specialized UAVs which do not follow general commercial trends. Sample size for each regression varies as the manufacturers do not consistently report all parameters; each regression therefore uses subset of database for which the parameters were available after outlier removal. The obtained relations are assessed for statistical error metrics to quantify their agreement with the dataset. The resulting exponents reflect observed design tendencies rather than strict physical laws. The derived scaling laws serve for the preliminary estimation of key design and performance parameters. The regression formulation is as used in the multirotor scaling literature[31], and does not seek to introduce a new mathematical model or improve regression accuracy per se. However, these scaling laws are then integrated into a sizing methodology, allowing early-stage designers to obtain preliminary configurations based on available constraints. Variables used are defined along with units in Sections 2.1-2.2 and Tables 1 and 2; subsequent equations use the same notation and units as defined therein, unless otherwise stated. For Class I, the plots display only the retained data points used for regression, and the sample sizes in Table 3 correspond directly to these. For Class II, the plots display the full database prior to outlier removal, while the sample sizes in Table 4 reflect only the subset retained for regression.

3.1. Scaling and Sizing of Class I multirotor

The class I multirotor database was utilized to develop empirical scaling laws between key design parameters, listed in Table 1. MTOW is taken as the primary variable, and each of the remaining parameters was individually regressed against it to form power-law relations. For all fitted relations of the form $y = \alpha x^{\beta}$, the 95% confidence intervals of (α,β), obtained via bootstrap resampling, are reported for both class I and class II in Tables 3 and 4 respectively. Fig. 4a illustrates the scaling trend between MTOW and endurance, represented by the regression fitted over 87 data points with a coefficient of determination $R^2 = 0.91$, including 95% confidence bounds. The endurance equation is given by

$$E_I = 0.5326 \cdot MTOW_I^{0.6325} \tag{4}$$

The data shows moderate scatter, particularly at lower MTOW, where design diversity is high. There's noticeable vertical scatter, some UAVs with the same MTOW differ by a significant factor in endurance. In contrast, the upper range exhibits tighter clustering, indicating similar endurance design practices among heavier platforms. This scaling relation can be used as a first-order estimate for endurance based on MTOW during conceptual trade-off studies. For example, the fitted curve

Table 3
Class I regression results ($y = \alpha x^{\beta}$) with 95% confidence intervals.

y	x	α	95% CI (α)	β	95% CI (β)	R^2	N
E_I	$MTOW_I$	0.5326	(0.415, 0.717)	0.6325	(0.569, 0.671)	0.9	87
BCap	$MTOW_I$	4.594	(4.11, 5.13)	1.025	(0.921, 1.119)	0.93	107
G	$MTOW_I$	41.013	(33.123, 44.969)	0.3234	(0.303, 0.358)	0.92	62
D_I	$MTOW_I$	0.639	(0.525, 0.766)	0.434	(0.400, 0.467)	0.94	58
L_I	G	1.041	(0.946, 1.306)	0.943	(0.890, 0.960)	0.95	158
B_I	G	0.77	(0.72, 0.83)	1	(0.84, 1.21)	0.95	149

Table 4
Class II regression results ($y = \alpha x^{\beta}$) with 95% confidence intervals.

y	x	α	95% CI (α)	β	95% CI (β)	R^2	N
WP_{II}	$MTOW_{II}$	0.3305	(0.1954, 0.4656)	1.0776	(0.9590, 1.1961)	0.92	52
WP_{II}	$MTOW_{II}$	0.7518	(0.1517, 1.1351)	0.8996	(0.7649, 1.0343)	0.92	21
AA_{II}	$MTOW_{II}$	0.1152	(0.018, 0.2493)	0.8396	(0.5182, 1.1610)	0.91	46
AA_{II}	$MTOW_{II}$	0.0191	(0.0031, 0.0352)	1.2254	(1.051, 1.399)	0.95	11
B_{II}	$MTOW_{II}$	0.2906	(0.249, 0.372)	0.5188	(0.453, 0.547)	0.91	60
L_{II}	$MTOW_{II}$	0.2851	(0.257, 0.364)	0.5198	(0.445, 0.529)	0.92	42
W_B	$MTOW_{II}$	0.447	(0.184, 0.710)	0.7648	(0.629, 0.900)	0.93	22
En_{II}	W_B	189.0712	(172.918, 205.223)	1.0376	(1.008, 1.066)	0.99	111

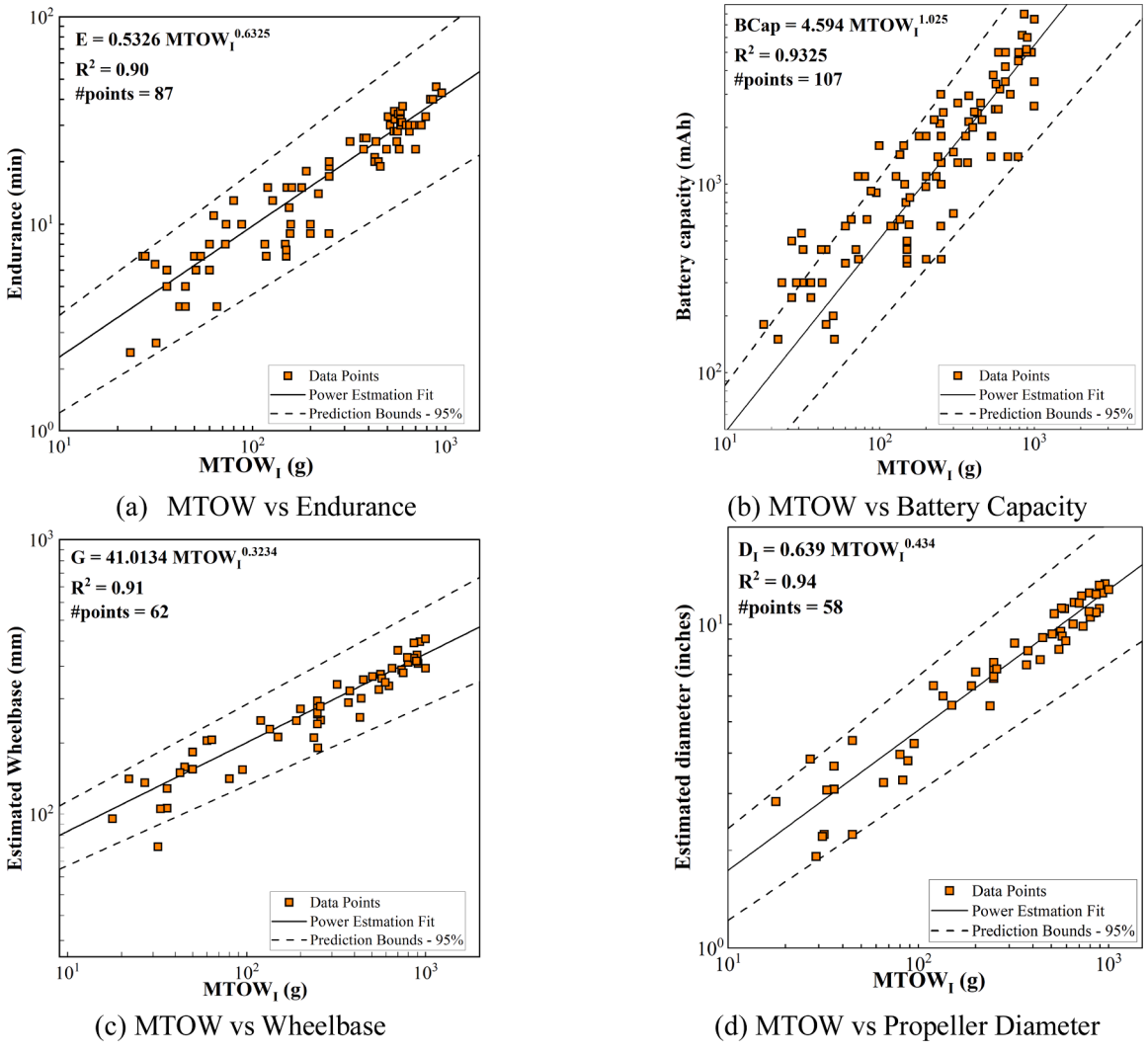


Fig. 4. Endurance, Battery Capacity, Wheelbase and Propeller Diameter of Class I Multirotor.

indicates that a typical 250 g commercial quadcopter has an endurance of about 17.5 minutes. Although drones such as the Syma Z6G[39] and XKJ K911 Max[40] approach this figure with endurance of 19 and 20 minutes, respectively, the Dji mini-series drones have an endurance of up to 30 to 35 minutes,[41] and the Husban Zino mini pro drone has an endurance of up to 40 minutes [42].

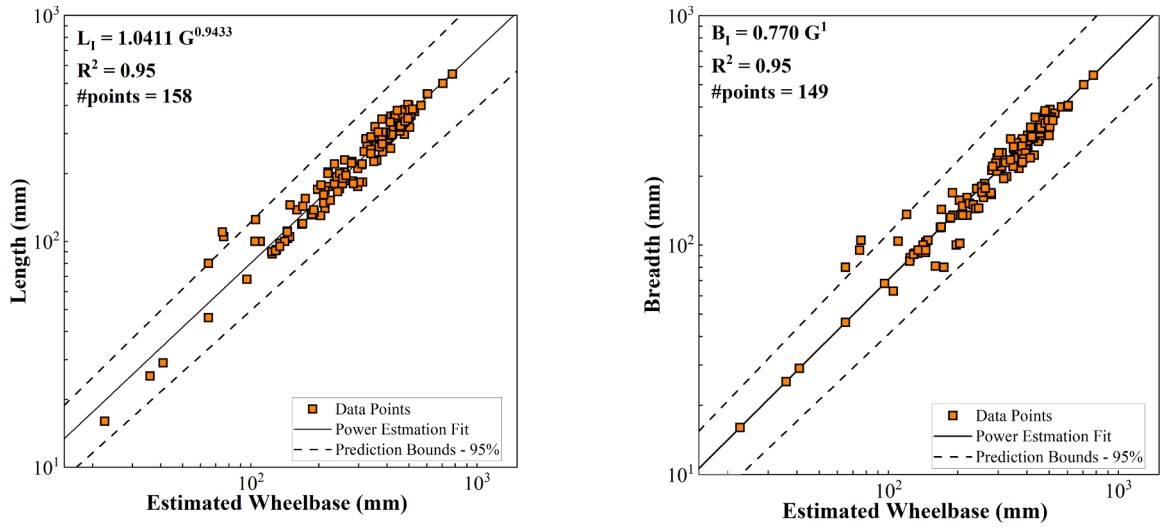
Similarly, Fig. 4b illustrates how MTOW and battery capacity are related, as described by the equation

$$BCap = 4.594 \cdot MTOW_I^{1.025} \tag{5}$$

with $R^2 = 0.9325$ over 107 samples. The exponent close to unity implies a near-linear scaling of battery capacity with MTOW, reflecting a proportional allocation of energy as vehicle size increases. For the same MTOW, several distinct battery capacity values can be observed. This variation arises due to intentional design-specific tradeoffs between compact recreational drones and high endurance photography drones. However, the overall narrow spread indicates that designers generally size battery capacity to achieve a target flight time, making it a tightly controlled parameter across the dataset for class I multirotors.

Fig. 4c and Fig. 4d show more subtle scaling behaviors of the dimensional parameters with respect to MTOW. The relationship between wheelbase (G) and MTOW is captured by the empirical relation,

$$G = 41.013 \cdot MTOW_I^{0.3234} \tag{6}$$



(a) Wheelbase vs Length (b) Wheelbase vs Breadth

Fig. 5. Length and Breadth of Class I Multirotor.

with a coefficient of determination $R^2 = 0.91$ based on 62 data points. The sub-linear exponent suggests that increases in vehicle mass is accompanied by proportionally smaller increases in structural span. Fig. 4c can be inferred that smaller drones vary more in frame size for a given weight, possibly due to mission diversity (indoor vs outdoor, racing vs camera drones). While, with an increase in MTOW, the distribution becomes denser and more aligned with the trend. In contrast, the propeller diameter (D) shows a moderately stronger dependency on MTOW, following the relation

$$D_I = 0.639 \cdot MTOW_I^{0.434} \tag{7}$$

with $R^2 = 0.90$ derived from 58 samples. The exponent suggests a sub-linear scaling, implying that while propeller size increases with MTOW, it does so at a slower rate than mass. This relationship is useful for estimating actuation area, disk loading, and thrust margin, especially when actual rotor data is unavailable. The equation tends to over-predict propeller diameter by approximately two inches across the range, owing to variations in frame clearance and hub spacing.

As length (L_I) and breadth (B_I) are found to correlate more directly with the wheelbase (G) as they are deterministically linked. They were regressed against G instead of MTOW, due to the geometric relationship that exists between these parameters in quadrotor layouts. Once the wheelbase is estimated from the MTOW through Equation. 6, then length and breadth can be obtained using the scaling relations as noted in Equation.8. This approach allows for a sequential sizing: starting from the MTOW, wheelbase is estimated, and then frame dimensions are inferred accordingly. Fig. 5a and Fig. 5b illustrate the regressions between the wheelbase and the overall dimensions of the vehicle such as length and breadth, based on 158 and 149 data points, respectively. The fitted relations indicate linear scaling trends.

$$L_I = 1.041 \cdot G^{0.943} B_I = 0.770 \cdot G^{1.0} \tag{8}$$

The linear exponent indicates that length and breadth scale linearly with wheelbase. Together, these regressions enable complete spatial estimation of the vehicle once MTOW is known, forming the

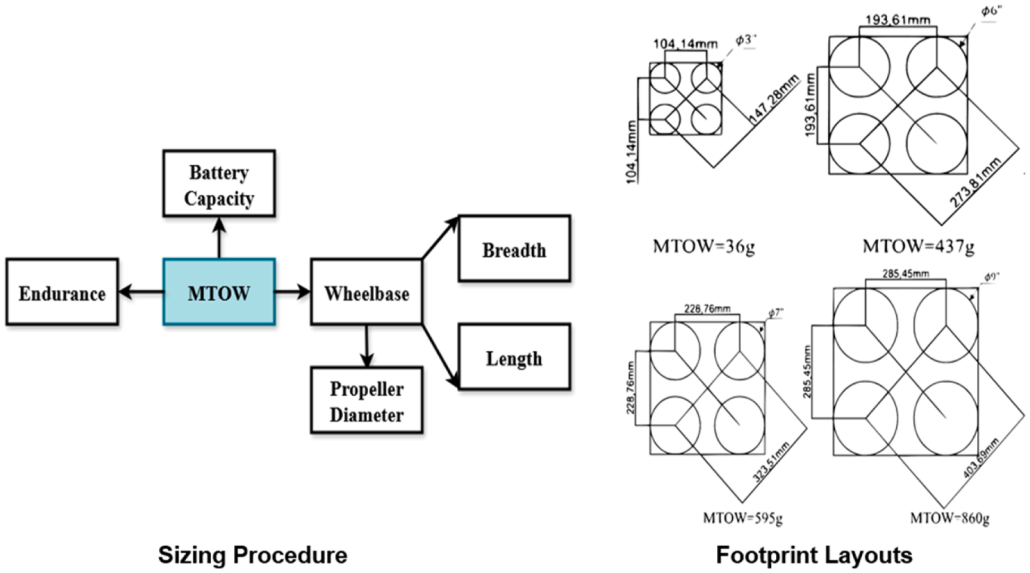


Fig. 6. Sizing Procedure and Schematic illustration representative 4-rotor footprint layouts with varying MTOW for class I.

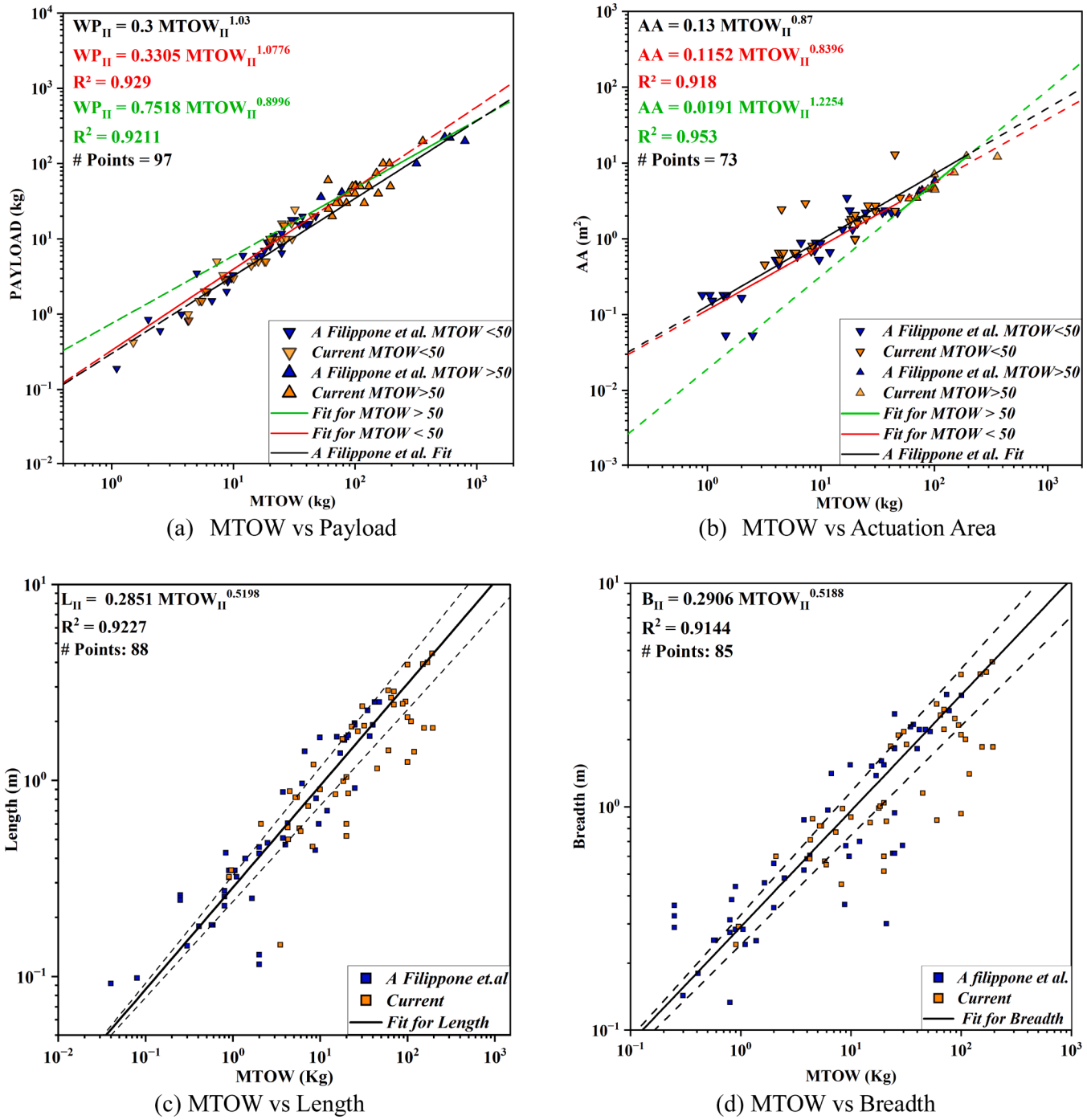
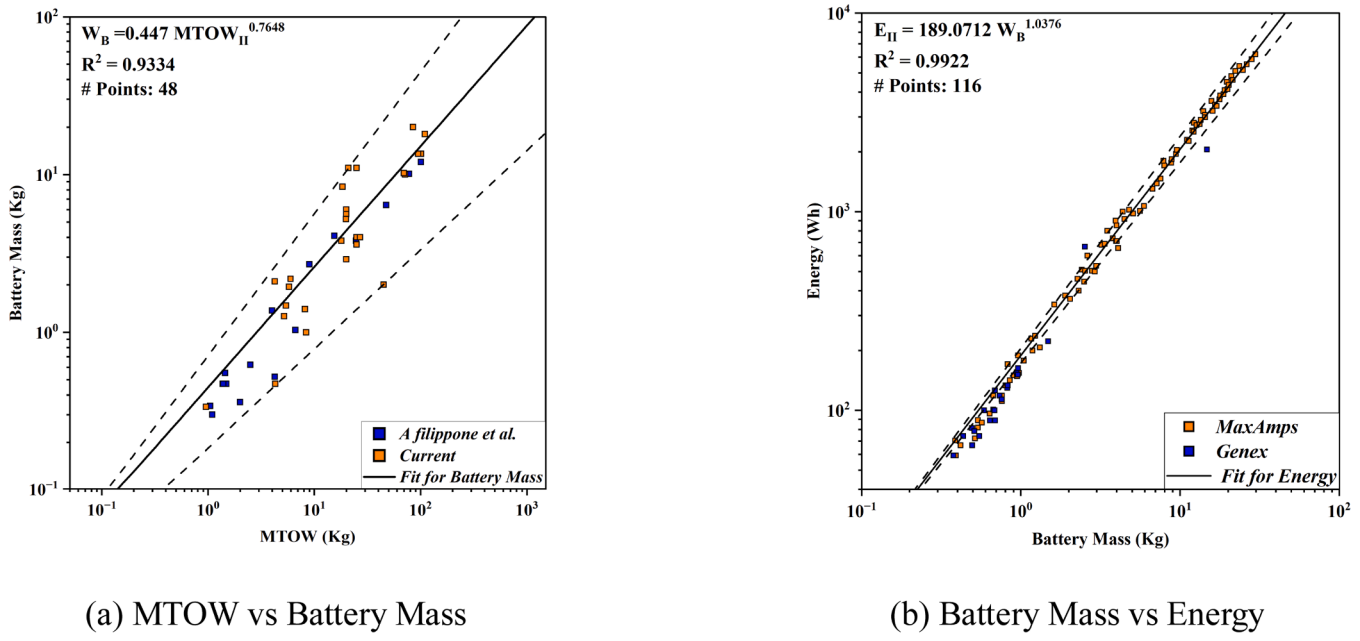


Fig. 7. Payload, Actuation Area, Length, and breadth of class II multirotor.

dimensional backbone of sizing methodology, as illustrated in Fig. 6. This structured methodology allows early-stage UAV designers to systematically progress from performance goals to geometric layout, i.e., sizing of the multirotor.

Initially, the wheelbase (G) is calculated using the single-variable power-law regression derived from scaling trends, Equation. 6. The battery capacity "E" is estimated from MTOW using the power-law relation in Equation.5, which aids in battery selection early in the design process. The length (L_I) and breadth (B_I) of the frame are derived from the wheelbase using Equation. 8. The propeller diameter (D_I) is estimated from the wheelbase using a geometric approximation outlined in Equation. 2. The estimated propeller diameter is used to calculate the actuation area (AA), which quantifies the aerodynamic footprint. Fig. 6 also illustrates four different class I UAVs sized using the proposed methodology. These are corresponding to MTOW's of 36 g, 437 g, 595 g, and 860 g, respectively. Table 3 provides a summary of the scaling laws and their statistical metrics, including coefficient of determination (R^2) and number of data points (N). Additionally, it should be noted that three performance-related parameters, wind resistance (V_{wI}), ceiling altitude (H_I), and maximum velocity (V_{maxI}), did not show statistically significant correlations with any of the other variables. This is expected due to the limited number of data available for these variables and the relatively narrow reported ranges. This observation aligns with existing literature, such as the study by Filippone et.al[31] which reports no discernible pattern between MTOW and maximum velocity for larger UAVs.



(a) MTOW vs Battery Mass (b) Battery Mass vs Energy

Fig. 8. Battery parameters for Class II multirotor.

3.2. Scaling and Sizing of Class II multirotor

The class II database was utilized to form scaling laws with the design variables listed in Table 2. Fig. 7a shows the scaling trend between MTOW and Payload mass. Payload mass is a crucial factor affecting multirotor design. The multirotor carries a wide variety of payloads, including LiDAR, thermal, and hyperspectral imaging, various other sensors depending on specific application, introduce significant mass and volume tradeoffs in the vehicle sizing process [43]. The plot exhibits a positive correlation; heavier platforms generally support greater payloads. However, the increase is not uniform across the range.

In the Fig. 7a, the blue line shows the fit for MTOW less than 50kg, while the green line shows the fit for MTOW greater than 50 kg. The black line represents the payload trend reported by Filippone et al. [31] The resulting scaling laws for payload mass WP_{II}, is expressed as a function of MTOW are:

$$WP_{II} = \begin{cases} 0.3305 \cdot MTOW_{II}^{1.0776} \text{if} & MTOW < 50kg \\ 0.7518 \cdot MTOW_{II}^{0.8996} \text{if} & MTOW > 50kg \end{cases} \quad (9)$$

Equation 9 describes the payload mass as a function of MTOW below and above 50kg. The trend for MTOW less than 50kg closely follows the existing trends in the literature, from Filippone et al. [31], both in terms of exponent and coefficient. In contrast, the MTOW greater than 50kg fit shows a distinct deviation, with a higher coefficient and a lower exponent. This implies that while heavier platforms have a larger payload fraction, the rate of increase in payload mass slows with an increase in MTOW. A nested model F test was conducted to confirm that introducing separate regressions below and above 50 kg significantly improves the MTOW-payload fit ($F(2,93) = 5.69$, $p = 0.0046$; $\Delta AIC = 7.2$).

Fig. 7b shows the scaling trend between the actuation area and the MTOW. The scaling equations are given by

$$AA_{II} = \begin{cases} 0.1152 \cdot MTOW_{II}^{0.8396} \text{if} & MTOW < 50kg \\ 0.0191 \cdot MTOW_{II}^{1.2254} \text{if} & MTOW > 50kg \end{cases} \quad (10)$$

In the lower MTOW range, as noted in Eq. 10, the exponent value of 0.8396 indicates that the actuation area grows steadily but at a slower rate relative to MTOW. This trend is consistent with the fit reported by Filippone et al. [31] However, above 50kg MTOW, the scaling behavior changes significantly, as noticed by the nature of the trend in Fig. 7b.

Fig. 7c and Fig. 7d present the relationships between MTOW with the overall physical footprint of the multirotor length and breadth of the multirotor, respectively. These empirical fits as given in Equation. 11 & Equation. 12 help to infer feasible geometric dimensions during early-stage design. For the length and breadth of the multirotors, the following regressions are observed:

$$L_{II} = 0.2851 \cdot MTOW_{II}^{0.5198} \quad (11)$$

$$B_{II} = 0.2906 \cdot MTOW_{II}^{0.5188} \quad (12)$$

Battery mass increases with MTOW, but at a less-than-proportional rate. This implies that as the multirotor size increases, the relative battery fraction decreases. This limits the endurance scalability of heavy UAVs despite increased capacity; power demand grows faster than stored energy. For design purposes, Equation 13 provides a first-order estimate of battery mass based on a given MTOW, helping to allocate weight fractions in early sizing.

Fig. 8a shows the relationship between MTOW and battery mass, captured by the scaling law:

$$W_B = 0.447 \cdot MTOW_{II}^{0.7648} \quad (13)$$

Fig. 8b presents the relationship between the battery mass (kg) and stored energy (Wh), based on the specifications of commercial MaxAmps [37] and GenX[38] lithium-polymer battery packs. The regression is given by:

$$En_{II} = 189.0712 \cdot W_B^{1.0376} \quad (14)$$

Together, these two regressions establish the relation between MTOW and available battery energy, through the battery mass. By combining Equation. 13 with Equation. 14, an MTOW-to-energy scaling can be inferred, Equation. 15.

$$En_{II} \propto MTOW_{II}^{0.7648 \times 1.0376} \approx MTOW_{II}^{0.7936} \quad (15)$$

Thus, the total energy available increases sub-linearly with MTOW, implying that endurance does not scale proportionally with size. This has implications for endurance and mission range, as energy supply does not keep pace with increasing system weight. Consequently, as vehicle size grows, improvements in efficiency, power management, or energy density become increasingly necessary to maintain or extend

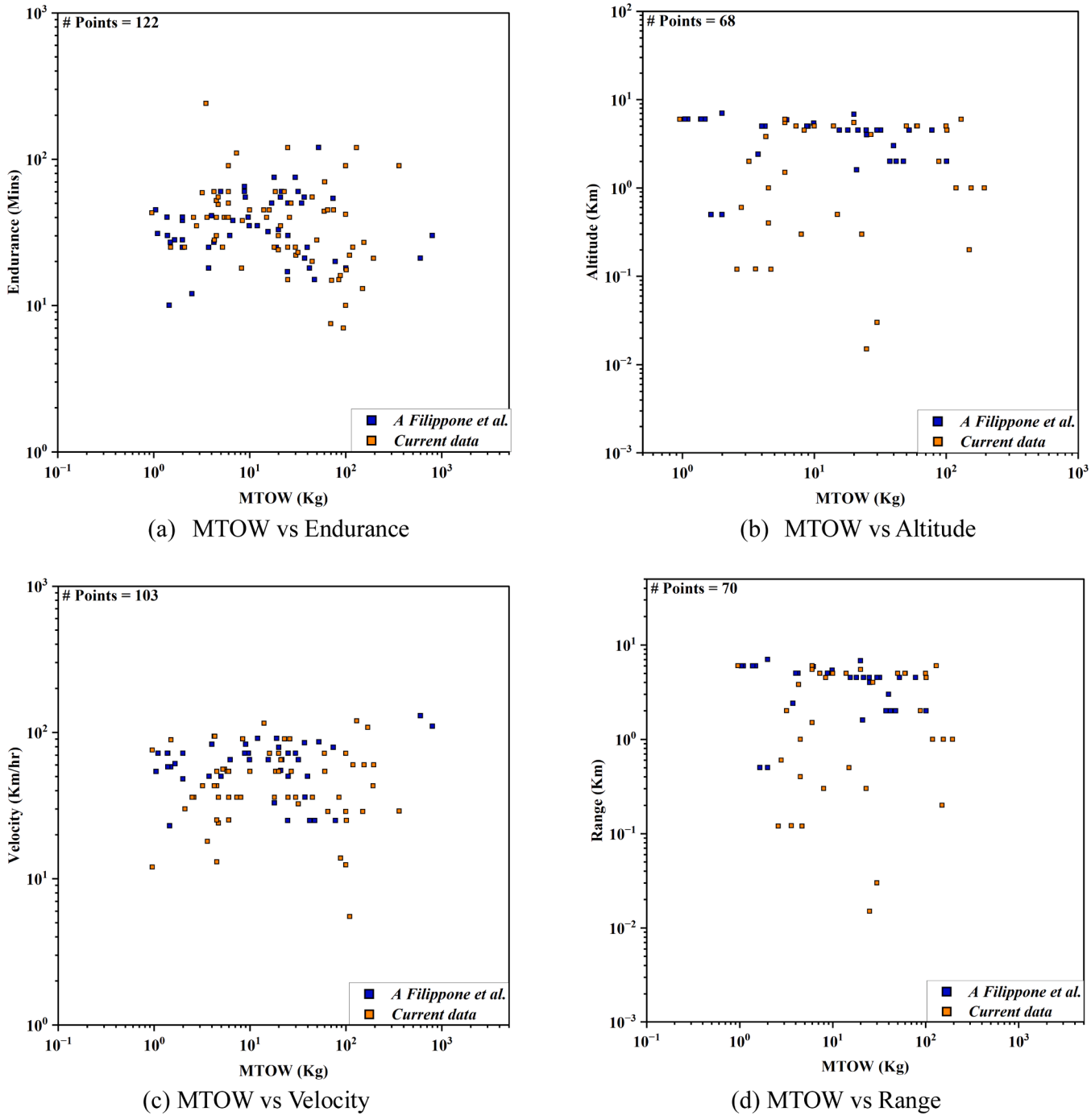


Fig. 9. Endurance, Altitude, Velocity, and Range of Class II multirotors.

performance. This is relevant for conceptual design, where endurance and range are functions of available energy. This reinforces how energy availability is inherently constrained by the MTOW, and in turn, how battery mass becomes a limiting design parameter in both small and large multirotors.

Table 5
Regression results for Eq. 16 with 95% confidence intervals and error metrics.

PARAMETER	ESTIMATE	95% CI	p-VALUE
R^2	0.977	-	-
ADJUSTED R^2	0.975	-	-
MAPE	14.19%	-	-
RMSPE	17.24%	-	-
N_p Exponent	-1.0258	(-1.238, -0.814)	< 0.001
W_p Exponent	0.7466	(0.676, 0.817)	< 0.001
AA Exponent	0.1123	(0.029, 0.195)	0.009

Fig. 9a- Fig. 9d present the observed relationships between MTOW and four key performance parameters: endurance, altitude, velocity, and range. Unlike the previously examined design metrics, these operational parameters do not exhibit meaningful correlation with MTOW. In Fig. 9a, the data points are widely scattered, and no clear monotonic trend is observed. The dispersion is expected as endurance is not solely determined by platform mass but is influenced heavily by mission profile, battery energy density, propulsion efficiency, and operational constraints such as payload or regulatory limits [44,45].

Fig. 9a and Fig. 9c, exhibit no strong functional relationship between

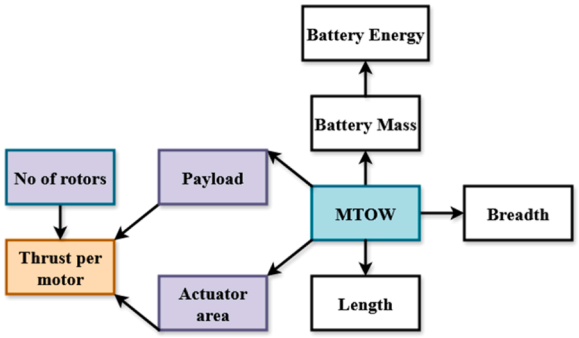


Fig. 10. Sizing Approach for Class II Multirotor.

or obtained via empirical fits from earlier equations. This allows for systems-level sizing based on a small set of input design requirements.

$$T = 4.3141 \cdot N_p^{-1.0258} \cdot AA_{II}^{0.1123} \cdot W_P^{0.7466} \quad (16)$$

The scaling laws and their performance metrics were listed in Table 4, which was used for a data-driven framework for early-stage sizing of Class II multirotors. Fig. 10 schematically depicts the methodology behind the sizing approach and shows the hierarchical relationship between the key variables used for the sizing. MTOW acts as the primary variable linking structural layout, actuator area, payload capacity, and battery mass. The actuator area, which governs disc loading and affects thrust generation and aerodynamic efficiency, is directly related to MTOW. The MTOW also dictates the multirotor's planform dimensions, length, and breadth, which in turn determines the multi-

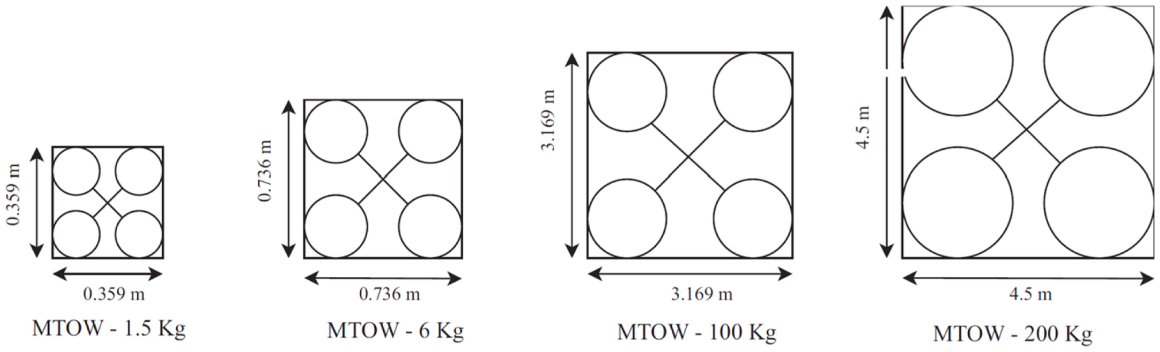


Fig. 11. Schematic illustration representative 4-rotor footprint layouts with varying MTOW for class II.

altitude & range with the MTOW. Different platforms have very different altitude performance; some lightweight systems can reach notable altitudes that are on par with or even higher than those of heavier platforms. The variation in velocity likely reflects differences in aerodynamic design, rotor disk loading, and intended operational role, some multirotors are optimized for hover stability and control, while others are designed for faster horizontal movement.

Similarly, Fig. 9d shows no strong relation between range and MTOW, which is attributable to design diversity. This observation is consistent with the findings of Bauersfeld and Scaramuzza[45], who demonstrated using a first-principles approach that multicopter range and endurance are governed by the interaction of aerodynamic, propulsive, electrical parameters, and characteristics of the chosen motor-propeller combination rather than MTOW alone [22,46].

Also, unlike linear one-to-one mappings, the scaling trends reflect the multivariate and interdependent nature of multirotor design space. For example, thrust per motor is not only a function of payload and number of rotors but also indirectly influenced by actuator area and MTOW through their effect on lift distribution and rotor efficiency. With the data set available, the study is extended in search of representing any design variable in terms of multiple other design parameters. To capture such relations, a multiple regression was conducted. Equation (16) is the multiple regression equation which relates the required hover thrust per motor to actuator area, payload mass, and rotor count. The regression was performed using N = 53 Class~II UAVs, drawn from the database described in Section 2.2, for which all three variables were simultaneously available; these UAVs are listed in supplementary material. Thrust per motor "T" is expressed in kgf, payload mass W_P in kg, actuator area AA in m^2, and N_p denotes the number of rotors. The equation represents a statistical trend rather than the physics-based model and should therefore be interpreted as giving first order estimates. The R^2 value, confidence interval, and error metrics in Table 5, indicate the expected uncertainty when using this relation. This relation estimates rotor thrust requirements once actuator area, payload, and rotor count are known, parameters that are either specified by mission requirements rotor's structural layout and physical footprint sizing. Additionally, payload mass is also constrained by MTOW.

In multirotor scaling, both the number of rotors and per-motor loading directly influence structural requirements, motor selection, and overall energy consumption. Per-motor thrust cannot be treated as an independent design variable, instead it arises from the coupled interaction among payload mass, number of rotors, actuator disk area, and the resulting maximum take-off weight (MTOW). The MTOW, in turn, defines the total lift requirement and constrains the feasible operating envelope of the rotors, indicating that the sizing process does not follow a simple unidirectional chain of dependencies. Furthermore, mission range or endurance requirements impose constraints on the battery capacity. An increase in battery capacity usually leads to higher battery mass, which inturn affects the MTOW and propagates back through the sizing loop. These interactions highlight the inherently coupled nature of multirotor sizing and performance optimization, where design variables are coupled.

Fig. 11 illustrates four different class II UAVs sized for various range of MTOW using the proposed route map as noted in Fig. 10. Therefore, the proposed scaling and sizing relations can be applied for the preliminary sizing of the multirotor depending on the MTOW requirement. It should be noted that MTOW need not be known a priori; when unavailable, the equations can be used in reverse by starting from a specified payload or mission requirement and estimating MTOW using the payload-MTOW relation, and then evaluate the remaining sizing parameters. A typical use case involves, selecting a target payload, estimating the corresponding MTOW using the regression for payload fraction, obtaining key geometric and power parameters such as rotor diameter, wheelbase, battery mass, and battery energy from the respective equations, and verifying that the resulting power loading and endurance lie within practical ranges. This provides a physically consistent way to size new configurations prior to detailed optimization or component selection. Unlike conventional non-statistical preliminary sizing approaches that start by assuming rotor dimensions or propulsion characteristics and proceed using actuator-disk or blade-element

Table 6
In-sample (IS) MAPE and repeated stratified 10-fold cross-validation (CV) results for scaling relations.

Class	Relation	IS MAPE (%)	CV MAPE (%)	CV SD (%)
I	MTOW vs Endurance	40.93	42.79	8.51
I	MTOW vs Battery Capacity	42.77	43.07	8.46
I	MTOW vs Wheelbase	22.52	25.01	10.51
I	MTOW vs Propeller Diameter	22.55	25.20	7.26
I	Wheelbase vs Length	18.56	23.41	8.74
I	Wheelbase vs Breadth	26.24	31.09	12.06
II	Payload (≤50 kg)	18.38	23.98	11.17
II	Payload (>50 kg)	21.69	23.67	8.70
II	Actuator Area (≤50 kg)	24.30	35.37	17.17
II	Actuator Area (>50 kg)	30.73	51.95	27.70
II	Battery Mass vs MTOW	31.61	35.52	11.81
II	Battery Energy vs Battery Mass	16.49	27.34	16.95
II	Length vs MTOW	28.78	32.45	8.84
II	Breadth vs MTOW	33.08	39.91	13.86

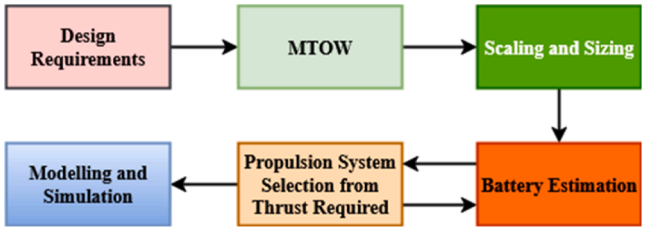


Fig. 12. Proposed Methodology for the complete sizing of multirotor.

Table 7
Design Specifications of the Heavy Lift Multirotor.

Parameter	Value	Unit
Payload Mass	60	kg
Estimated MTOW	130.09	kg
Number of Rotors	8	rotor
Rotor Diameter	1.09	m
Estimated Battery Mass	18.5	kg
Estimated Length	3.5	m
Estimated Breadth	3.6	m
Configuration	Coaxial	-

formulations or powertrain component selection methods, the present framework has inverse approach. It provides a feasible overall UAV scale and parameters using empirically derived MTOW-based scaling relations. This gives a configuration-independent estimate for early-stage design, reducing dependency on unavailable aerodynamic or motor-performance data. Physics based analysis can later improve the design in higher design stages.

To check whether the scaling relations generalise beyond the UAVs used for regression, a repeated stratified k-fold cross-validation (10-fold, 100 repetitions) was performed. The database was split into 10 folds such that each fold contains a similar distribution of MTOW values. In each run, the model was trained on 90% of the data and evaluated on the remaining 10%, 100 such runs were conducted. The regression approach follows as described in Section 3, with power-law fitting performed in log-log space, and 2σ outlier removal applied only within the training folds of each split, such that test-fold data do not influence outlier boundaries. The folds are stratified by MTOW so that each fold contains a representative distribution of small and large UAVs. The mean cross-validation MAPE and its standard deviation are reported in Table 6. For most relations, the cross-validation error is close to the in-sample error. This indicates that the relations are not overfitting and give consistent predictions on unseen data. Higher deviation and variability are observed in some cases, especially for actuator area and battery energy. This is due to limited data and greater variation in designs. Overall, the scaling relations generalise well within the range of commercially available UAVs represented in the dataset.

4. Statistical Design of a Heavy-Lift Multirotor- A Case Study

In this section, a case study is presented to demonstrate the complete sizing methodology, component selection for a class II multirotor, in particular, a heavy lift multirotor along with the dynamics and control of the vehicle. Fig. 12 illustrates the design methodology proposed for a class II multirotor. Heavy lift multirotors have been the focus of recent studies on design, stability, and sizing, reflecting their growing role in large payload aerial operations [2,12,47–51], such as construction, disaster relief, forestry, agriculture, and defense logistics. To demonstrate the proposed design methodology, the payload mass of 60 kg is adopted as a primary design criterion. This value represents typical heavy-load payloads in scenarios such as pesticide spraying, timber transport, humanitarian aid delivery, and construction material handling, and serves as a scalable baseline for this study of the statistical design of the multirotor.

The multirotor under consideration, with a payload of 60 kg, falls into the MTOW category greater than 50 kg, as indicated in Fig. 7a. The estimated design specifications are noted in the Table 7, derived from the established regression models. Using Equation. 9, the corresponding MTOW is calculated to be 130.09 kg. The actuator area is calculated from Equation. 10, and a preliminary bounding box is drawn as noted in Fig. 13a based on scaling laws from Equation. 11 and Equation. 12. This serves as the basis for preliminary UAV sizing, with projected

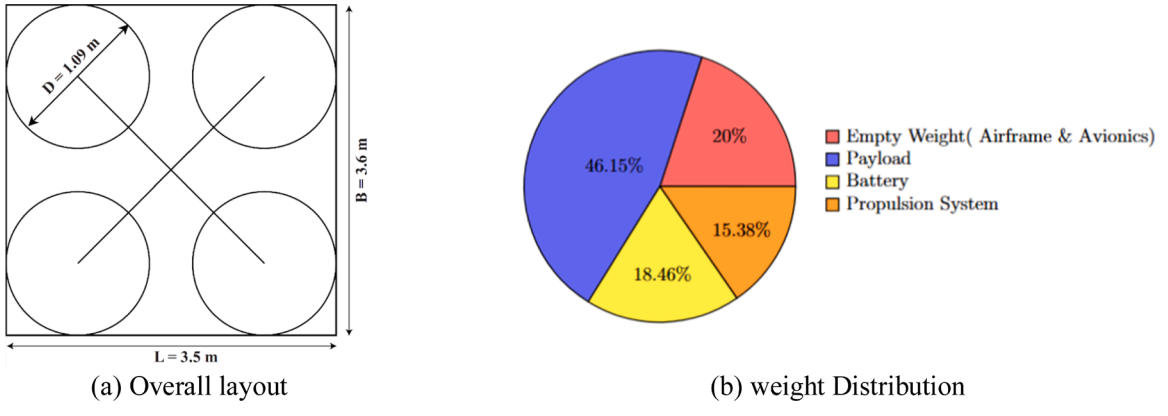


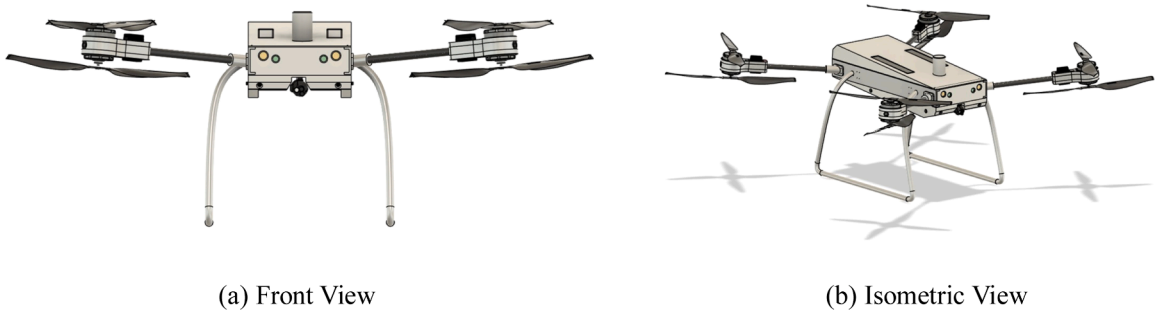
Fig. 13. Layout and Weight distribution of 60 kg payload Co-axial Heavy Lift Multirotor.

Table 8
Propulsion System.

Part Name	Quantity	Weight (kg)	Total (kg)
Tiger Motor U15II 80KV[54]	8	1.74	13.92
G40*13.1 Prop[55]	8	0.237	1.896
Thunder 1224S 200A ESC[56]	8	0.62	4.96
LiPo Battery 46000mAH 15S2P 55.5v[57]	2	12.43	24.86

match the energy requirement of the motor. The selected battery should match the energy specification of the motor, rather than the battery mass.

Based on these design specifications, the propulsion components were selected and listed in Table 8 along with the individual component mass. It can be noted that two compatible batteries can match the energy requirement, but with increased mass from the initial estimate. The weight split for the propulsion system is listed in Table 8, and the weight



(a) Front View (b) Isometric View

Fig. 14. Proposed model of Co-axial Heavy Lift Multirotor.

dimensions of 3.5 m in length and 3.6 m in breadth. The rotor diameter of 1.09 m is estimated using Equation. 3 and Equation 10. To fit the multirotor design in the estimated size of length and breadth, an eight-rotor coaxial configuration was selected, based on established design practices and its frequent use in UAVs across similar MTOW ranges [12,51,52]. Coaxial configurations provide a favorable thrust-to-volume ratio and inherent torque cancellation, beneficial for all-terrain operations and urban navigation. The gain in compactness comes at the expense of aerodynamic efficiency [12].

Energy requirements are estimated using the MTOW-battery mass relation from Equation. 13 and the available energy from Equation. 15, resulting in an initial battery mass of approximately 18 kg. The thrust required per motor is calculated using the multiple regression in Eq.16, with inputs $N_p = 8$, AA $= 7.44\ m^2$ (from Eq.10 at MTOW $= 130.09$ kg), and $W_P = 60$ kg; the Eq.16 output is 13.6 kgf per motor. To account for take-off and off-hover power requirements, this is multiplied by a safety factor of 1.5, giving a design thrust of 20.4 kgf per motor. In general, to account for excess power, thrust per motor is multiplied by a factor of 1.5 to deliver the power requirement other than hover [60,61]. The total thrust for 8 motors is 163.4 kgf, giving total thrust-to-weight ratio of 1.256. Recent experimental work shows that for a given propeller diameter and motor speed, thrust and torque behave predictably while efficiency still varies significantly with battery and motor choice, highlighting the importance of including propulsion efficiency in sizing relations [44]. Considering that motor efficiency peaks at 60-75 % throttle,[53] knowing the battery mass and battery capacity, a commercially available battery was identified by repeated search to

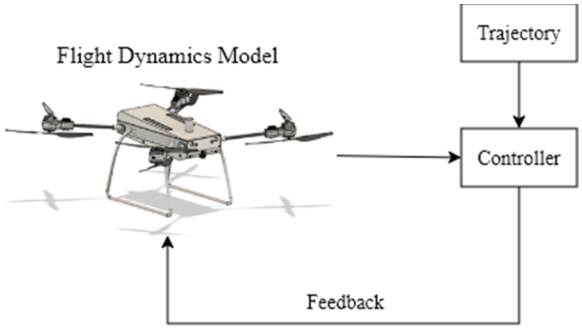


Fig. 15. Dynamics and Controller Model.

percentage is noted in Fig. 13b. The representative CAD models of the heavy lift multirotor corresponding to these specifications are illustrated in Fig. 14a and Fig. 14b.

Unlike fixed-wing aircraft, rotorcraft lack a simple and universally applicable analytical framework for evaluating inherent static stability [59]. In order to validate the design and complete the multirotor system development, a Software-In-the-Loop Simulation (SILS) is performed in MATLAB/Simulink. Such Software-in-the-loop simulation validation follows standard practices in multirotor research, as surveyed by Pekša and Mamchur,[58] who emphasize the importance of virtual testing environments for evaluating control responsiveness before field trials.. The parameters like mass (m) and the moment of inertia tensor (J), are obtained from the CAD models. These parameters are then fed into the multirotor's dynamic equations of motion as noted by Quan,[36] forming a complete mathematical representation of the system dynamics. Subsequently, a suitable PID based control system is adopted to study the dynamics response of the designed system through simulation, and ensure the stable flight of the proposed design. PID architectures are the standard approach for attitude and position control in multirotor literature, due to their simplicity and proven effectiveness [47–49,58]. The controller evaluates the system response to representative flight trajectories, assessing the dynamic stability and control characteristics of the designed UAV.

Fig. 15 provides a schematic representation of the dynamics and controller model. A cascade PID control structure, consisting of inner and outer loops, is implemented to control the multirotor dynamics. Six PID controllers are implemented to control the states x, y, z, θ, ϕ, ψ, with the attitude controllers (θ, ψ) cascaded within the position controllers (x, y). In order to assess the reliability of the controller, it is tested against a set of representative trajectories, which are listed in Table 9. The selected trajectories include circular, conical spiral, and leminiscate paths. These paths have varying spatial complexity and dynamic demands, which test the system's performance in position tracking along the X, Y, and Z axes. The simulation results for the trajectories given in

Table 9
Trajectory Parameters.

Trajectory	X(t)	Y(t)	Z(t)
Circular	2·sin (0.1t)	2 - 2·cos (0.1t)	10
Conical Spiral	(2πt/50)·sin (0.1t)	(2πt/50) - (2πt/50)·cos (0.1t)	0.5t
Lemniscate	10·sin (0.1πt)	10·cos (0.1πt)	0.5t

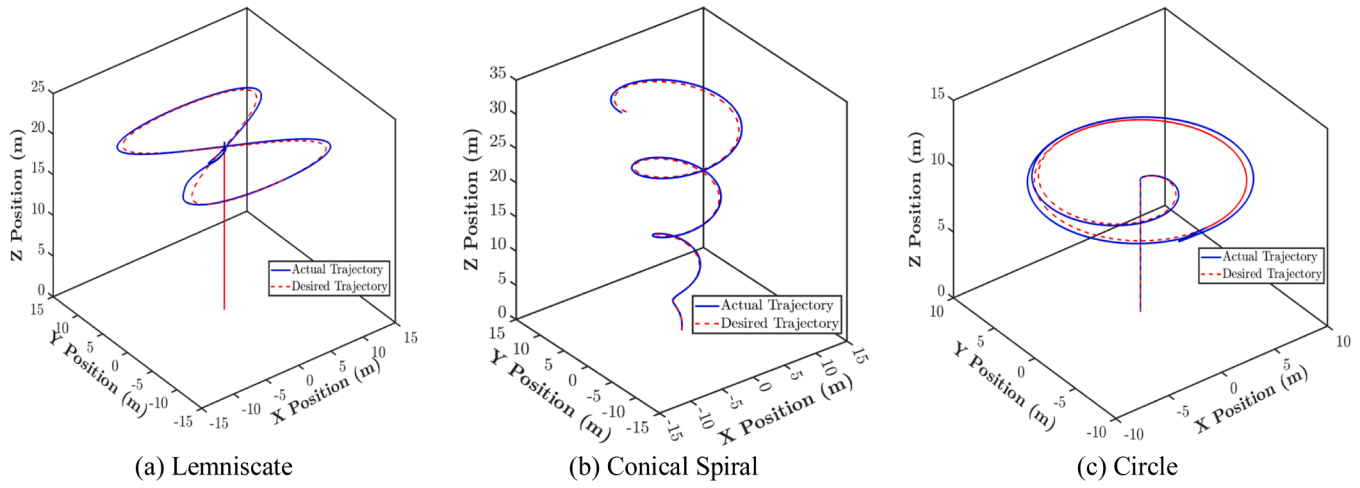


Fig. 16. Path shapes: Lemniscate, Conical Spiral, and Circle.

Table 10
Class II, Predicted vs. actual values.

UAV	MTOW (kg)	Payload (Kg)		Length (m)		Breadth (m)		Batt. Mass (kg)	
		Actual	Predicted	Actual	Predicted	Actual	Predicted	Actual	Predicted
D230A	3.5	–	1.27	0.14	0.54	0.07	0.55	–	1.16
FLIR Skyranger	5.0	3.5	1.87	–	0.65	–	0.67	–	1.53
JMRRC	32	24.5	13.84	1.9	1.72	1.9	1.75	–	6.33
HF C50 Agri	95	50.0	45.21	2.52	3.04	2.31	3.08	13.5	14.55
Gryphon GD-120X	119	30.0	55.37	1.4	3.42	1.4	3.47	–	17.28

Table 11
Class I, Predicted vs. actual values.

UAV	Type	MTOW (g)	Endurance (min)		Wheelbase (mm)		Prop. Dia. (in)	
			Actual	Predicted	Actual	Predicted	Actual	Predicted
TizzyToy BL01	TD	453.6	40.0	25.5	–	296.5	–	9.09
Lumenier Defender	AP	785	4.0	36.1	457	354.1	5.2	11.53
tznflydrone DH 120	TD	450	7.0	25.4	–	295.8	–	9.06
QAV-PRO Mirage	FPV	660	5.0	32.3	–	334.8	4.0	10.7
Pi G1 Vlogger	AP	1000	12.0	42.1	–	382.9	–	12.81
iFlight BumbleBee HD	FPV	539.6	–	28.5	145	313.7	3.0	9.8
Hummingbird V3 BNF	TD	23.3	2.4	3.9	–	113.6	1.22	2.51
AOS RC V5	FPV	1000	–	42.1	220	382.9	5.1	12.81
AcroBee75 HD O3	FPV	99	3.5	9.7	75	181.3	1.57	4.69
Walkera F2103D	RD	370	5.0	22.4	–	277.6	–	8.32

Table 9 are shown in Fig. 16. The effectiveness of the proposed controller in precisely controlling the multirotor's position is ensured by the low tracking errors seen across all paths.

Multirotor design space is inherently large, which has multiple feasible designs, depending on mission priorities. The scaling relations derived in this study represent the dominant trends observed in commercially available UAVs, such as agricultural, imaging, and general purpose UAVs, and therefore provide one representative design point within this broader design space. Deviations from the predicted values are expected when UAVs are designed with objectives that differ from these dominant trends, such as UAVs optimised for thrust and agility, compact coaxial heavy-lift systems, or designs with unusually high payload fractions.

To clarify applicability, the sizing relations are applied to representative UAVs, with predicted parameters compared against actual values, and are reported in Tables 10 and 11 for Class II and Class I, respectively. For Class II, some UAVs show deviations from the scaling trends due to differences in design preferences and configuration. The JMRRC has a higher payload fraction than the general commercial trend, leading to under-prediction by the framework. The D230A and FLIR Skyranger are designed for surveillance, having different mass distribution with compact structural layouts, resulting in deviations in both payload and geometry. The Gryphon GD-120X, a coaxial heavy-lift UAV, achieves the required thrust through rotor stacking, resulting in a smaller planform than predicted by trends. While, the HF∼C50 agricultural UAV shows good agreement, as its design aligns with the general commercial trends captured in the dataset.

For Class∼I, FPV and racing UAVs (e.g., QAV-PRO Mirage, iFlight BumbleBee, AOS RC V5, AcroBee75, Walkera F210) prioritise thrust-to-weight ratio and agility over endurance. This results in smaller frame sizes and propeller diameters, as well as lower endurance than predicted by scaling relations. Similarly, the Lumenier Defender shows reduced endurance and an oversized predicted geometry due to its design emphasis. Endurance optimised UAVs such as the TizzyToy BL01 exceed predicted endurance. These deviations are expected and occur due to different mission priorities, propulsion configurations, and design

strategies. The framework gives a reliable first-order estimate corresponding to common design trends, while deviations occur when designs follow different objectives within the broader design space and require refinement beyond the initial estimates.

5. Conclusion

This work has attempted to address a long-standing lacuna in multirotor UAV design by using a statistical, class-aware methodology for preliminary sizing. In contrast to the prevailing reliance on fragmented heuristics and rule-of-thumb approximations, this study develops a framework based on one of the more comprehensive datasets of commercial multirotors collected to date. The study avoids adopting a single general rule for all 309 multirotors by dividing them into two classes according to their maximum take-off weight. The results are summarized as follow:

- Prominent clustering in the sub-250 g range is observed in Class I database, which is a result of market preference for nano-UAVs. On the other hand, the Class II database contains heavy-lift UAVs weighing more than 50 kg, owing to growing industrial demand.
- In Class I multirotors, it is observed that geometric parameters like length and breadth scale more consistently with wheelbase than they do with MTOW directly. From the collected database of both classes, no correlations could be derived between MTOW and performance metrics like wind resistance, ceiling altitude, and velocity.
- Distinct scaling behaviour is noted below-50 kg and above-50 kg systems listed in Class II database. Payload mass varies nearly linear with MTOW below 50 kg whereas for MTOW above 50 kg variation is sub-linear. The sub-linear nature indicates diminishing payload fractions in the heavy-lift domain. For actuator area, the point-estimate exponent rises from 0.84 to 1.23 across the 50 kg threshold.
- Collectively, findings demonstrate that the preliminary sizing requires class sensitive regression and the universal scaling exponents are insufficient and potentially misleading when used across wide MTOW ranges.
- From the heavy-lift 60 kg payload platform case study, it is demonstrated that statistical relations give reasonable estimates of rotor dimensions, battery mass, and subsystem requirements. The estimates obtained are comparable to those of commercially available components. The dynamic characteristics of the statistically sized vehicle was estimated using MATLAB/Simulink. The sized UAV successfully followed three different flight trajectories demonstrating dynamic stability of the UAV. This supports the feasibility of the proposed sizing methodology and highlights the importance of inculcating dynamic simulation into the design process of the multirotor.

By presenting first-order estimates for rotor dimensions, structural span, and energy requirements, it allows designers to avoid ad hoc thumb-rules. Additionally, the analysis identifies weight regimes where simple power law models fail to provide better predictions due to shifts in payload and rotor efficiency. Performance parameters like velocity, range, and service ceiling are not sufficiently correlated with MTOW. Since this dataset is restricted to commercial platforms (excluding military prototypes), statistical imbalance is expected. Development of extensive open-source database to include all multirotor configurations would further improve the industrial relevance of the framework. Multirotor UAV design is inherently multivariate; this study provides statistical based preliminary scaling relations to determine feasible UAV scale from existing multirotors. The framework provides first order estimates for the preliminary design, scaling relations derived as univariate regressions; the design is complemented by physics-based analyses or optimization in subsequent stages, where mission-specific trade-offs are refined, and further serve as inputs to multivariate or mission-driven optimization, representing a promising direction for future work extending this framework.

CRediT authorship contribution statement

Aditya Thokur Ravi: Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Manan V Jain:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Formal analysis, Data curation. **Sashank Erukala:** Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Vallabhu Narayana Murthy:** Writing – review & editing, Writing – original draft, Visualization, Resources, Investigation, Formal analysis, Data curation. **Vishwa Karthi:** Visualization, Software, Investigation, Formal analysis, Data curation. **Narendiran V:** Visualization, Investigation. **Rajesh Senthil Kumar T:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.rineng.2026.111398.

Appendix A: Sensitivity of Regression Results to Outlier Removal Criterion

To make sure that the scaling relations are not sensitive to the choice of outlier removal criterion, the regression was rerun for: (i) no outlier removal, (ii) 3σ removal, and (iii) 2σ removal. The resulting coefficients (α) and exponents (β) for both Class I and Class II are presented in Tables A1 and A2, respectively.

Table A1
Regression coefficient (α) and exponent (β) to outlier removal criterion for class I.

Relation	No outlier removal		3σ		2σ	
	α	β	α	β	α	β
MTOW vs Endurance	2.021	0.402	1.760	0.422	0.533	0.633
MTOW vs Battery capacity	17.04	0.814	16.91	0.810	4.594	1.025
MTOW vs Wheelbase	114.9	0.188	117.5	0.183	41.01	0.323

(*continued on next page*)

Table A1 (*continued*)

Relation	No outlier removal		3σ		2σ	
	α	β	α	β	α	β
MTOW vs Propeller dia.	2.665	0.188	2.726	0.183	0.639	0.434
Wheelbase vs Length	0.830	0.978	0.830	0.978	1.041	0.943
Wheelbase vs Breadth	0.530	1.041	0.530	1.041	0.770	1.000

Table A2
Regression coefficient (α) and exponent (β) to outlier removal criterion for class II.

Relation	No outlier removal		3σ		2σ	
	α	β	α	β	α	β
Payload (<50 kg)	0.303	1.115	0.340	1.077	0.331	1.078
Payload (>50 kg)	1.264	0.792	1.190	0.795	0.752	0.900
Actuation area (<50 kg)	0.129	0.908	0.271	0.631	0.115	0.840
Actuation area (>50 kg)	0.196	0.720	0.219	0.685	0.019	1.225
Battery mass	0.500	0.728	0.449	0.740	0.447	0.765
Battery mass vs Energy	189.8	1.036	194.5	1.030	189.1	1.038
Length	0.325	0.465	0.304	0.475	0.285	0.520
Breadth	0.483	0.390	0.347	0.432	0.291	0.519

Data availability

We have uploaded our database as supplementary files.

References

[1] Kumar Vijay, Michael Nathan, Opportunities and challenges with autonomous micro aerial vehicles, Int J Robot Res 31 (11) (2012) 1279–1291, https://doi.org/10.1177/0278364912455954.

[2] Kotarski Denis, Piljek Petar, Kasać Josip, Heavy-Lift Multirotor Unmanned Aerial Vehicles Characterization and Sizing, IEEE Access 11 (2023) 94290–94305, https://doi.org/10.1109/ACCESS.2023.3310820.

[3] Bouabdallah Samir, Marcelo Becker, Siegwart Roland, Autonomous miniature flying robots: coming soon! - Research, Development, and Results, IEEE Robot Autom Mag 14 (3) (2007) 88–98, https://doi.org/10.1109/MRA.2007.901323.

[4] Lim Daejin, Kim Hyeongseok, Yee Kwanjung, Mission-oriented performance assessment and optimization of electric multirotors, Aerosp Sci Technol 115 (2021) 106773, https://doi.org/10.1016/j.ast.2021.106773.

[5] Raymer Daniel, Aircraft Design: A Conceptual Approach, Sixth Edition, American Institute of Aeronautics and Astronautics, Inc., Washington, DC, 2018.

[6] Hassanalian Mostafa, Khaki Hamed, Khosravi Mehrdad, A new method for design of fixed wing micro air vehicle, Proc Inst Mech Eng Part G J Aerosp Eng 229 (5) (2015) 837–850, https://doi.org/10.1177/0954410014540621.

[7] Verstraete Dries, Palmer Jennifer L, Hornung Mirko, Preliminary Sizing Correlations for Fixed-Wing Unmanned Aerial Vehicle Characteristics, J Aircr 55 (2) (2018) 715–726, https://doi.org/10.2514/1.C034199.

[8] A. Aboelezz, M. Hassanalian, A. Desoki, B. Elhadidi, G. El-Bayoumi, Design, experimental investigation, and nonlinear flight dynamics with atmospheric disturbances of a fixed-wing micro air vehicle, Aerosp Sci Technol 97 (2020) 105636, https://doi.org/10.1016/j.ast.2019.105636.

[9] L. de Angelis Emanuele, Giulietti Fabrizio, Rossetti Gianluca, Bellani Gabriele, Performance analysis and optimal sizing of electric multirotors, Aerosp Sci Technol 118 (2021) 107057, https://doi.org/10.1016/j.ast.2021.107057.

[10] Winslow Justin, Hrishikeshavan Vikram, Chopra Inderjit, Design Methodology for Small-Scale Unmanned Quadrotors, J Aircr 55 (3) (2018) 1062–1070, https://doi.org/10.2514/1.C034483.

[11] M. Budinger, A. Reysset, A. Ochotorena, S. Delbecq, Scaling laws and similarity models for the preliminary design of multirotor drones, Aerosp Sci Technol 98 (2020) 105658, https://doi.org/10.1016/j.ast.2019.105658.

[12] Ong Wayne, Srigrarom Spot, Henrik. Hesse, Design Methodology for Heavy-Lift Unmanned Aerial Vehicles with Coaxial Rotors, AIAA Scitech 2019 Forum, San Diego, California, 2019.

[13] Dutta Abhishek, Robust design of a multirotor aerial vehicle, Sci Rep 11 (1) (2021) 21137, https://doi.org/10.1038/s41598-021-00413-4.

[14] K P Peeyush, V. Akshith, D. Sai Dakshesh, M V S Krishna Pranav, T. Malavika Menon, Development of Agricultural Drone for Crop Monitoring and Disease Detection, in: IEEE Int. Conf. Electron. Comput. Commun. Technol 2024, CONECCT, Bangalore, India, 2024, pp. 1–6.

[15] M. Hima, Thampatty K.C. Sindhu, Solar Powered Unmanned Aerial Vehicle. 20237th Int, in: Conf. Comput. Appl. Electr. Eng.-Recent Adv., CERA, Roorkee, India, 2023, pp. 1–6.

[16] Manalathody Abhishek, Kishoor Krishnan, Subramanian Santhana, Thangavel Jisnu Adindtra, Pushpangathan Sakthivel, V. Jinraj, Paranjothy Hariprasanth, Nonlinear and Linear PID Controllers-Based Hybrid Flight Control Strategy for a Quadcopter With Slung Load, IEEE Access 13 (2025) 48520–48542, https://doi.org/10.1109/ACCESS.2025.3551622.

[17] Vemulapalli Nehith Sai, Paladugula Pruthvi, Prabhat Guggilam Sai, Don S. Reinforcement Learning-Based Autonomous Landing of AirSim Simulated Quadcopter in Unreal Engine. 202415th Int, in: Conf. Comput. Commun. Netw. Technol. ICCCNT, Kamand, India, 2024, pp. 1–7.

[18] Kulkarni Eeshan, Sundaram Suresh, Design and Development of Scissorbot: A Novel Mid-Flight-Span-Reducing Quadcopter, J Mech Robot 17 (6) (2025) 061003, https://doi.org/10.1115/1.4067068.

[19] Shahmoradi Javad, Roghanchi Pedram, Hassanalian Mostafa. Design, analysis and prototyping of a spherical drone for underground mines, Int J Theor Appl Multiscale Mech 4 (1) (2022) 58, https://doi.org/10.1504/IJTAMM.2022.122895.

[20] Hassanalian Mostafa, Conceptual design, bioinspiration, and multidisciplinary analysis of drones, New Mexico State University, Las Cruces, New Mexico, USA, 2018.

[21] M. Hassanalian, A. Abdelkefi, Classifications, applications, and design challenges of drones: A review, Prog Aerosp Sci 91 (2017) 99–131, https://doi.org/10.1016/j.paerosci.2017.04.003.

[22] Biczyski Marcin, Sehab Rabia, Whidborne James F, Krebs Guillaume, Luk Patrick, Multirotor Sizing Methodology with Flight Time Estimation, J Adv Transp 2020 (2020) 1–14, https://doi.org/10.1155/2020/9689604.

[23] Bershadsky Dmitry, Haviland Steve, Johnson Eric N. Electric Multirotor UAV Propulsion System Sizing for Performance Prediction and Design Optimization. 57th AIAAASCEAHSASC Struct. Struct. Dyn. Mater. Conf.; San Diego, California, USA; 2016. p.

[24] S. Delbecq, M. Budinger, A. Ochotorena, A. Reysset, F. Defay, Efficient sizing and optimization of multirotor drones based on scaling laws and similarity models, Aerosp Sci Technol 102 (2020) 105873, https://doi.org/10.1016/j.ast.2020.105873.

[25] D. Lorenz Ralph, Flight Power Scaling of Airplanes, Airships, and Helicopters: Application to Planetary Exploration, J Aircr 38 (2) (2001) 208–214, https://doi.org/10.2514/2.2769.

[26] Omri Rand, Rand Omri, Vladimir Khromov, Khromov Vladimir, Helicopter Sizing by Statistics, J Am Helicopter Soc 49 (3) (2004) 300–317, https://doi.org/10.4050/jahs.49.300.

[27] Filippone Antonio, Data and performances of selected aircraft and rotorcraft, Prog Aerosp Sci 36 (8) (2000) 629–654, https://doi.org/10.1016/S0376-0421(00)00011-7.

[28] Raja Vijayanandh, Solaiappan Senthil Kumar, Rajendran Parvathy, Madasamy Senthil Kumar, Jung Sunghun, Conceptual Design and Multi-Disciplinary Computational Investigations of Multirotor Unmanned Aerial Vehicle for Environmental Applications, Appl Sci 11 (18) (2021) 8364, https://doi.org/10.3390/app11188364.

[29] P. Pounds, R. Mahony, Design principles of large quadrotors for practical applications, in: 2009 IEEE Int. Conf. Robot. Autom., Kobe, 2009, pp. 3265–3270.

[30] Avanzini Giulio, L. De Angelis Emanuele, Giulietti Fabrizio, Optimal performance and sizing of a battery-powered aircraft, Aerosp Sci Technol 59 (2016) 132–144, https://doi.org/10.1016/j.ast.2016.10.015.

[31] Ardelean Iuliu-Cezar, Usov Dima, Filippone Antonio, Parslew Ben, Hollingsworth Peter, Unmanned Wingless Rotorcraft Scaling, J Aircr 60 (5) (2023) 1553–1563, https://doi.org/10.2514/1.C037299.

[32] C. Ampatis, E. Papadopoulos, Parametric Design and Optimization of Multi-Rotor Aerial Vehicles, editor, in: J. Daras Nicholas (Ed.), Appl. Math. Inform. Sci. Eng., Appl. Math. Inform. Sci. Eng., 91, Springer International Publishing, Cham, 2014, pp. 1–25.
[33] Gatti Mauro, Giulietti Fabrizio, Preliminary Design Analysis Methodology for Electric Multirotor, IFAC Proc Vol 46 (30) (2013) 58–63, https://doi.org/10.3182/20131120-3-FR-4045.00038.
[34] Nam Taewoo, Soban Danielle, Mavris Dimitri. A Generalized Aircraft Sizing Method and Application to Electric Aircraft. 3rd Int. Energy Convers. Eng. Conf.; San Francisco, California; 2005.
[35] Gur Ohad, Rosen Aviv, Optimizing Electric Propulsion Systems for Unmanned Aerial Vehicles, J Aircr 46 (4) (2009) 1340–1353, https://doi.org/10.2514/1.41027.
[36] Quan Quan, Introduction to Multicopter Design and Control, Springer Singapore, Singapore, 2017.
[37] LiPo Batteries - USA - MaxAmps Lithium Batteries. Available at https://maxamps.com/. Accessed December 7, 2025, n.d.
[38] GenX Power Premium Lipo Battery: Robokits India, Easy to use, Versatile Robotics & DIY kits. Available at https://robokits.co.in/batteries-chargers/drone-batteries/genx-power-premium-lipo-battery. Accessed December 7, 2025, n.d.
[39] SYMA. SYMA Official Site. SYMA Official. Available at https://www.symatoys.com/. Accessed December 6, 2025, n.d.
[40] K911 max Drone - 8K Dual Camera 360 Laser Obstacle Avoidance. RCDrone. Available at https://rcdrone.top/products/k911-max-drone. Accessed December 17, 2025, n.d.
[41] DJI - Official Website. DJI Official. Available at https://www.dji.com/global. Accessed December 7, 2025, n.d.
[42] hubsan.com/na/. Available at https://www.hubsan.com/na/. Accessed December 17, 2025, n.d.
[43] Ganesh Kumar Siva Sivamani, Gudipalli Abhishek, A comprehensive review on payloads of unmanned aerial vehicle, Egypt J Remote Sens Space Sci 27 (4) (2024) 637–644, https://doi.org/10.1016/j.ejrs.2024.08.001.
[44] Goli Srikanth, Kurtuluş Dilek Funda, M. Alhems Luai, M. Memon Azhar, Hamda Imran Imil, Experimental study on efficient propulsion system for multicopter UAV design applications, Results Eng 20 (2023) 101555, https://doi.org/10.1016/j.rineng.2023.101555.
[45] Bauersfeld Leonard, Scaramuzza Davide, Range, Endurance, and Optimal Speed Estimates for Multicopters, IEEE Robot Autom Lett 7 (2) (2022) 2953–2960, https://doi.org/10.1109/LRA.2022.3145063.
[46] W. Prouty Raymond, Helicopter performance, stability, and control. Repr. with corr, Krieger Publ. Co, Malabar, Florida, 1995.
[47] Kotarski Denis, Piljek Petar, Pranjić Marko, Kasać Josip, Concept of a Modular Multirotor Heavy Lift Unmanned Aerial Vehicle Platform, Aerospace 10 (6) (2023) 528, https://doi.org/10.3390/aerospace10060528.
[48] Hamandi Mahmoud, Tognon Marco, Franchi Antonio, Direct Acceleration Feedback Control of Quadrotor Aerial Vehicles, in: 2020 IEEE Int. Conf. Robot. Autom, ICRA, Paris, France, 2020, pp. 5335–5341.
[49] Suprapto Bhakti Yudho, Heryanto M Ary, Suprijono Herwin, Muliadi Jemie, Kusumoputro Benyamin, Design and development of heavy-lift hexacopter for heavy payload. 2017 Int. Semin. Appl. Technol. Inf. Commun, ISemantic, Semarang, 2017, pp. 242–247.
[50] Ozdogan Gokhan, Leblebicioglu Kemal, Design, Modeling, and Control Allocation of a Heavy-Lift Aerial Vehicle Consisting of Large Fixed Rotors and Small Tiltrotors, IEEEASME Trans Mechatron 27 (5) (2022) 4011–4021, https://doi.org/10.1109/TMECH.2022.3150713.
[51] Turkmen Abdullah, Altug Erdinc, Design of a Quad-Jet VTOL UAS for Heavy-lift Applications, in: 2020 Int. Conf. Unmanned Aircr. Syst, ICUAS, Athens, Greece, 2020, pp. 875–882.
[52] Mishra Aadya, Pal Sourav, Singh Prabhat, Design and analysis of an Eight Rotor Co-Axial UAV using carbon fiber composites, Mater Today Proc 68 (2022) 1011–1015, https://doi.org/10.1016/j.matpr.2022.08.206.
[53] US Department of Energy (USDOE). "Determining Electric Motor Load and Efficiency.", Jan. 1997.
[54] T. Motor, U15II KV80 Heavy-lift UAV /Drone Motor - 36kg Max.thrust, T MotorT-MOTOR U15 V2 KV80 Motor U-Power - High Efficiency Heavy-Lift UAV Motor (December 6, 2025). Available at, https://store.tmotor.com/product/u15-v2-motor-u-power-kv80.html. Accessedn.d.
[55] T Motor Propeller. G40*13.1 UAV Propellers - Carbon Fiber Glossy | T-Motor Store. Available at https://store.tmotor.com/product/g40x13_1-prop-2pcs-1pair-glossy-carbon-fiber.html. Accessed December 6, 2025, n.d.
[56] Thunder 24S 200A ESC for High Power UAVs /Drones. Available at https://store.tmotor.com/product/thunder-200a-esc.html. Accessed December 6, 2025, n.d.
[57] LiPo 46000mAh 15S2P 55.5v Battery Pack - MaxAmps Lithium Batteries. Available at https://maxamps.com/collections/15s-lipo-battery-55-5v/products/lipo-46000-15s-55-5v-battery-pack. Accessed December 7, 2025, n.d.
[58] Peksa Janis, Mamchur Dmytro, A Review on the State of the Art in Copter Drones and Flight Control Systems, Sensors 24 (11) (2024) 3349, https://doi.org/10.3390/s24113349.
[59] D. Padfield Gareth, Helicopter Flight Dynamics Including a Treatment of Tiltrotor Aircraft, 3rd ed., John Wiley & Sons, 2018.
[60] A. Datta, *Introduction to helicopter and tiltrotor conceptual design* (NASA/CR-20210000284), National Aeronautics and Space Administration, 2021.
[61] W. Johnson, Rotorcraft Aeromechanics, Cambridge University Press, Cambridge, 2013.
[62] Federal Aviation Administration, Unmanned Aircraft Systems (UAS) Registration Task Force (RTF) Aviation Rulemaking Committee (ARC) final report, November 21, U.S. Department of Transportation, 2015, https://www.faa.gov/sites/faa.gov/files/uas/resources/policy_library/RTF-ARC-Final-Report.pdf.